

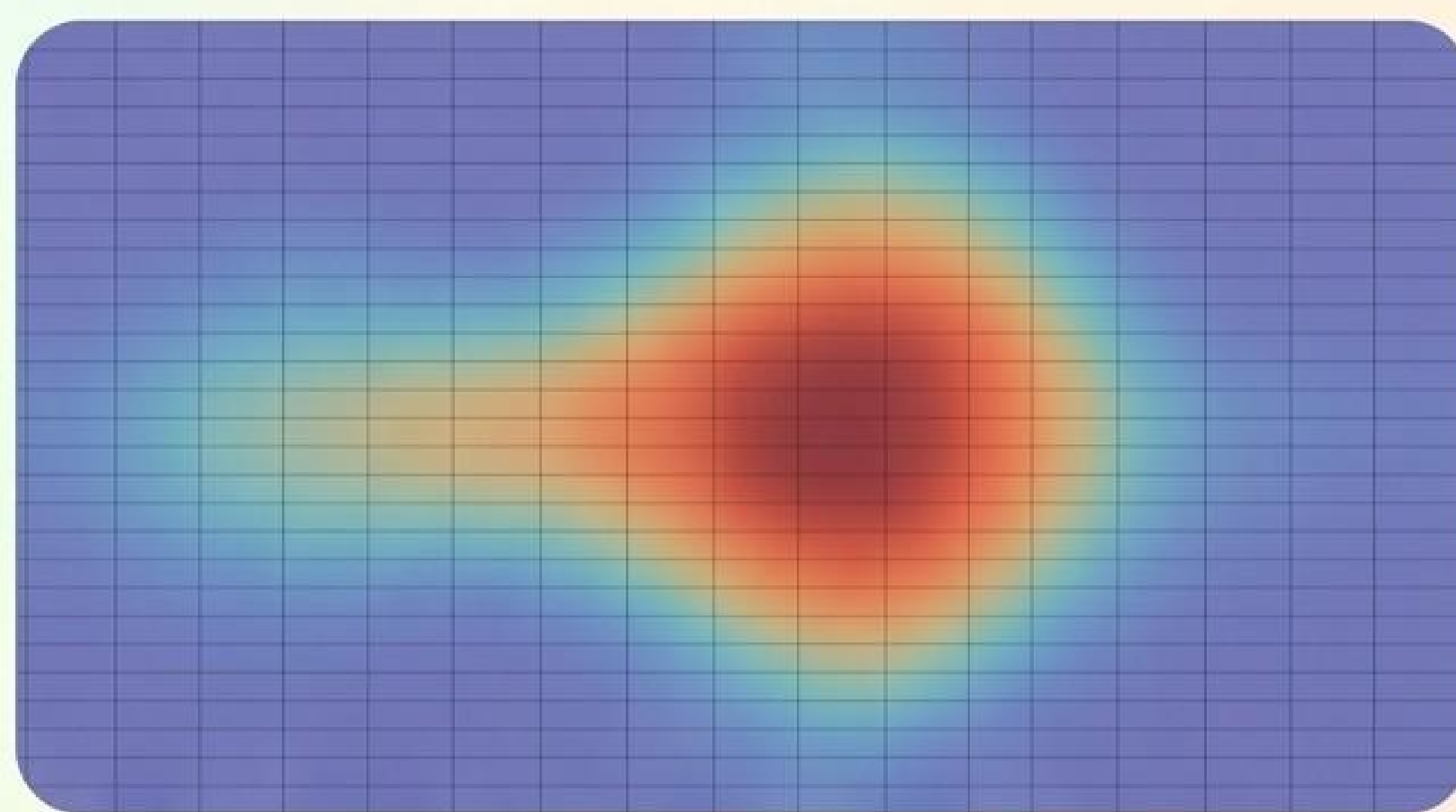
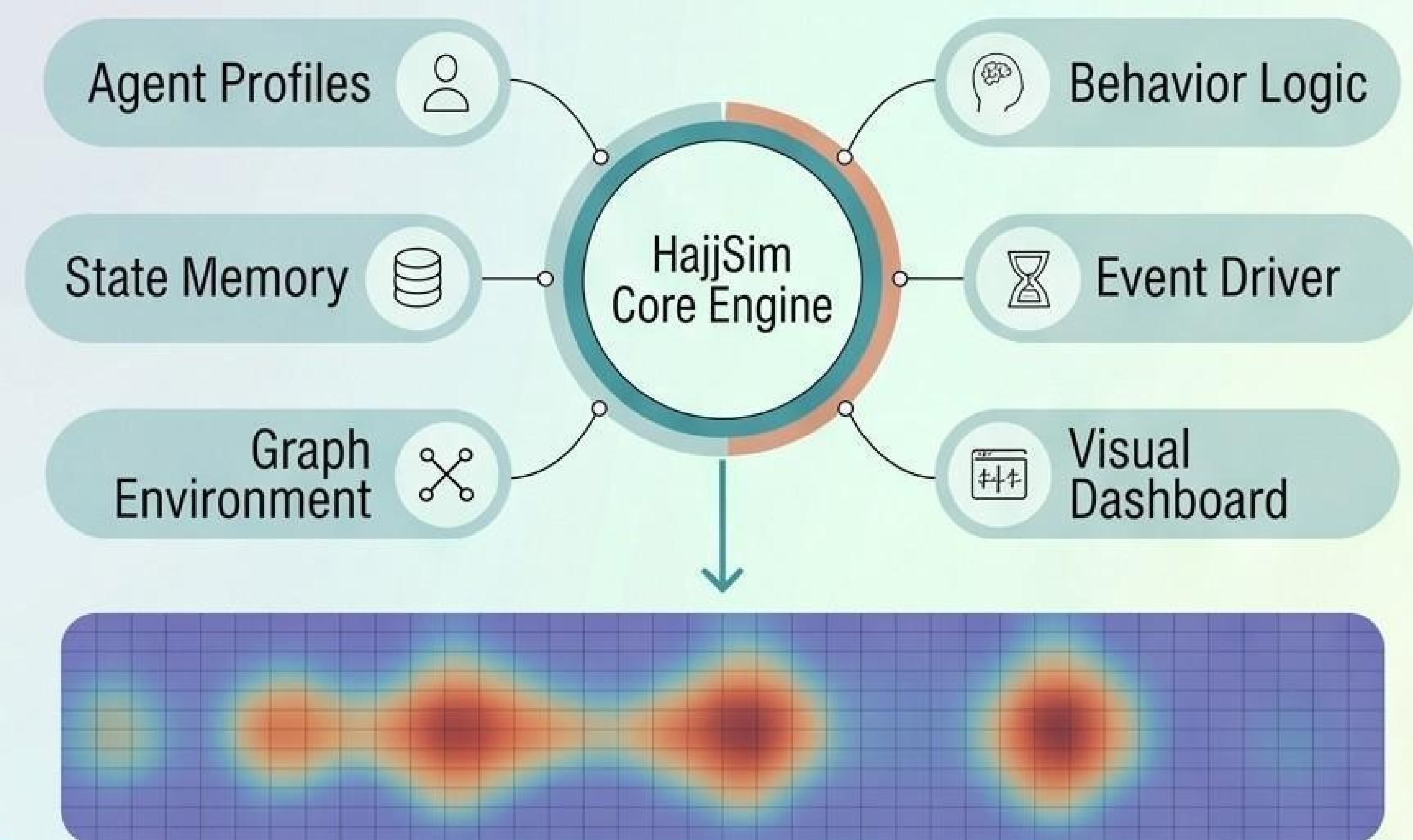
## Abstract

This project presents a predictive simulation platform that models Hajj crowd movement using multi-agent swarm intelligence. Each **pilgrim** is represented as an autonomous agent to simulate and identify potential risks before they occur.

## Introduction

Managing large-scale crowds during Hajj is highly complex due to extreme density and time constraints. Traditional systems rely on reactive monitoring, which may not prevent dangerous situations in advance.

## Project Description – Digital Twin Architecture



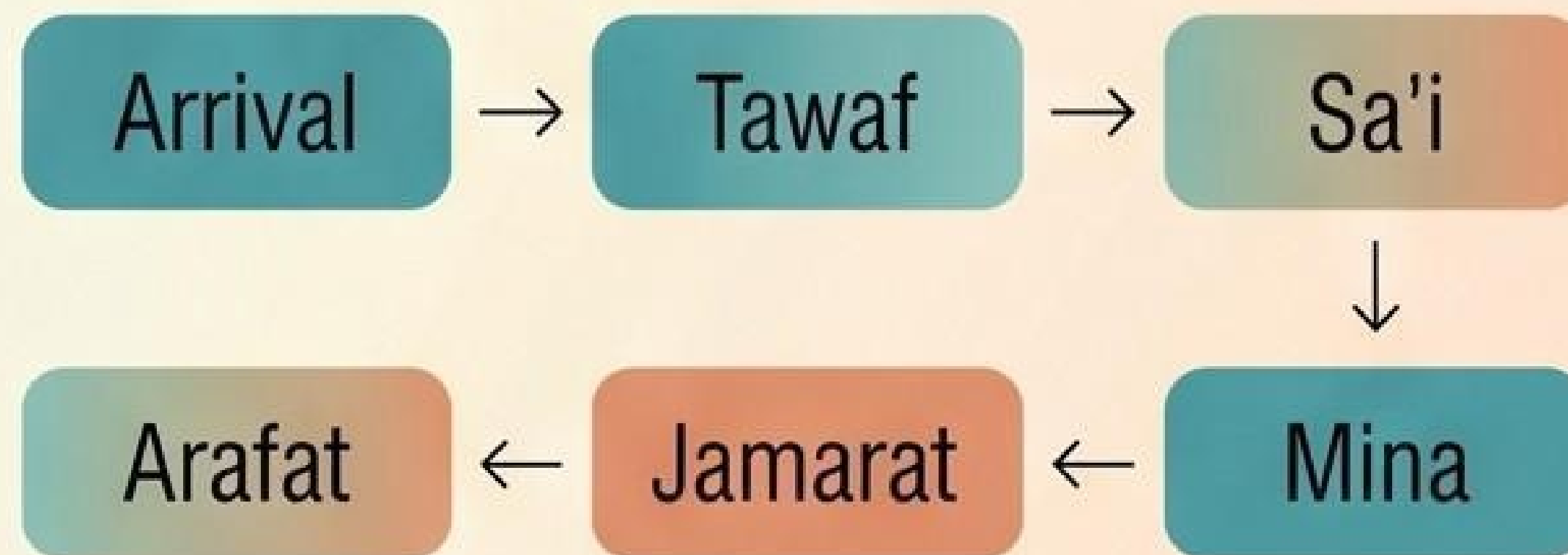
**Crowd Density Heatmap - Jamarat Bridge Simulation**

Thousands of autonomous agents simulate pilgrim movement across ritual stages, tracking stress, density, and hazard propagation in real time.

## Methods

- Agent-Based Modeling platform. Each agent includes:
  - Profile
  - State
  - Memory
  - Behavior Logic
- Environment modeled as graph-based spatial structure.
- Simulation runs using event-driven staged execution.

## Movement Flow Diagram



## Results & Conclusions

The system successfully demonstrates the ability to predict crowd stress and detect potential bottlenecks before they become critical. It supports a shift from reactive crowd to proactive safety management.



## Key System Features

- Real-time crowd density tracking
- Hazard injection & propagation modeling
- Visual heatmap dashboard
- Proactive safety bottleneck detection

## References

- [1] Bonabeau et al. (1999), Swarm Intelligence: From Natural to Artificial Systems.
- [2] Helbing & Molnar (1995), Social Force Model for Pedestrian Dynamics, Physical Review E.
- [3] Macal & North (2010). Tutorial on Agent-Based Modelling and Simulation. Journal of Simulation.